Fine-Tuning Does Not Correct: Abductive Accumulation and the Persistence of Prior Frames in Neural Systems

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Abstract

The standard account of fine-tuning treats it as correction: a new training distribution overwrites prior model behavior and installs updated representations. We argue this account is structurally incorrect. Drawing on a theory of *abductive accumulation* [**?**], we propose that fine-tuning is better modeled as *frame archival*: prior interpretive frames are not deleted from weight space but archived alongside the new frame, with a modified selection mechanism determining which frame is activated at inference time. Under this model, prior-frame reactivation is not a failure of fine-tuning but a structural consequence of how neural systems implement belief revision. This has direct implications for hallucination persistence, bias rebound after RLHF, and the design of debiasing interventions. We identify three testable predictions distinguishing the accumulation model from the standard correction model, and propose mechanistic interpretability as the primary empirical route for verification.

1 Introduction

Large language models (LLMs) exhibit a well-documented failure mode: behaviors suppressed by fine-tuning or reinforcement learning from human feedback (RLHF) reappear under distributional shift, adversarial prompting, or sufficiently novel contexts [**?**, **?**, **?**]. The standard explanation is incomplete fine-tuning: the corrective training signal was insufficient to fully overwrite the prior behavior. The prescription follows directly: more data, more compute, stronger regularization.

We propose a different diagnosis. The reappearance of suppressed behaviors is not a failure of correction. It is the expected behavior of a system that *archives* rather than *overwrites* prior interpretive frames. Under the accumulation model we develop here, fine-tuning installs a new frame at the top of an interpretive stack while the prior frame is retained in weight space with a modified selection probability. Suppressed behaviors reappear

precisely because the prior frame is reactivated under conditions that resemble its original acquisition context—not because correction was incomplete, but because correction, in the relevant sense, did not occur.

This reframing has practical consequences. If the accumulation model is correct, increasing the amount of corrective fine-tuning data will not eliminate prior-frame reactivation; it will reduce the selection probability of the prior frame under typical conditions while leaving the reactivation channel open. Reliable debiasing requires instead a modification of the *selection mechanism* that suppresses prior-frame activation, not merely stronger overwriting of the prior frame content.

Remark 1 (Terminological note on "overwrite"). *See [?] for the precise definition. In this paper,* overwrite *denotes a frame-dominant transition (superposition + selection-dominance), not deletion of prior frames.*

2 The Accumulation Model

2.1 Interpretive Frames and Overwrite

We adopt the formal framework of [?,?]. An *interpretive frame* $F(R)$ is a function from inputs to outputs induced by the representational state R of a system. For an LLM, R corresponds to the weight configuration; $F(R)$ is the conditional distribution $P(\text{output} \mid \text{input}, R)$.

The standard correction model assumes that fine-tuning implements an *overwrite transition* $R_t \to R_{t+1}$ in which the new frame $F(R_{t+1})$ replaces $F(R_t)$:

$$\text{(Overwrite):} \quad R_{t+1} = \arg\min_R \mathcal{L}(R, D_{\text{ft}})$$

where D_{ft} is the fine-tuning dataset and prior behavior is treated as eliminated up to the residual of the optimization.

2.2 Accumulation as the Default Mechanism

We propose instead that neural learning implements *accumulation*: the prior frame $F(R_t)$ is archived in weight space rather than deleted, and the new frame $F(R_{t+1})$ is superimposed onto the same weight space.

Definition 1 (Interpretive Stack in Weight Space). *The weight configuration R of a fine-tuned model implicitly encodes a layered interpretive stack $\mathcal{L} = (F_{t+1}, F_t, F_{t-1}, \ldots)$, where F_{t+1} is the most recently installed frame and earlier frames are encoded as residual structure in the weight space. The effective output frame at inference time is determined by a* selection rule *$\mathcal{S}$ that maps the input context e to an index k, activating frame F_k.*

Definition 2 (Selection Rule). *A selection rule $\mathcal{S} : \mathcal{E} \to \{1, \ldots, n\}$ maps the current input context e—including prompt structure, distributional features, and contextual similarity to prior training conditions—to an index k selecting the active frame. A candidate functional form is:*

$$\mathcal{S}(e) = \arg\max_k \ \text{softmax}_k\big(s(e, C_k)/\tau\big)$$

where C_k is the acquisition context of frame F_k, $\delta(\cdot,\cdot)$ is a context-similarity metric on $\mathcal{E}$, and $\tau > 0$ is a temperature parameter.

Remark 2. *This is not merely a metaphor. Mechanistic interpretability research has identified that LLMs encode competing "circuits"—distinct computational subgraphs responsible for different behaviors [?, ?]. Fine-tuning modifies the relative activation probability of these circuits without deleting the prior circuits. The accumulation model provides the theoretical framework within which this empirical phenomenon is the expected outcome rather than an anomaly.*

2.3 The Non-Correctability Proposition

Proposition 1 (Prior Frame Persistence)**.** *Let R_{t+1} be the weight configuration after fine-tuning on D_{ft}, and let R_t encode the prior frame F_t. Under the accumulation model, F_t remains present in R_{t+1} as a residual structure with acquisition context C_t. The reactivation probability of F_t satisfies:*

$$P\big(\mathcal{S}(e) = k \mid F_k = F_t\big) > 0 \quad \textit{whenever } s(e, C_t) > \epsilon$$

for some threshold $\epsilon > 0$ determined by the selection temperature τ. This bound is non-zero regardless of the amount of fine-tuning data in D_{ft}, because $\mathcal{S}$ responds to input-context similarity to C_t, which is a property of the input distribution rather than of the corrective training signal.

Corollary 1. *Increasing the volume of corrective fine-tuning data reduces the default selection probability of F_t (by shifting $\mathcal{S}$ toward F_{t+1} across the typical input distribution) but does not eliminate the reactivation channel. Prior-frame reactivation under out-of-distribution inputs is therefore a structural consequence of accumulation, not a failure of sufficiency.*

3 Distinguishing Predictions

The accumulation model and the standard correction model make distinct empirical predictions. We identify three that are tractable with current mechanistic interpretability tools.

3.1 Prediction 1: Context-Dependent Reactivation

Correction model: Suppressed behavior should be absent regardless of input context (up to residual optimization error, which should be uniformly distributed over the input space).

Accumulation model: Reactivation probability is highest for inputs e with large $s(e, C_t)$—that is, inputs that resemble the acquisition context of the prior frame. Suppression failures should be systematically concentrated in regions of input space close to the original pretraining distribution.

Test: Measure the rate of suppressed behavior reappearance as a function of the distributional distance between the test input and the fine-tuning distribution. The accumulation model predicts a monotonically increasing reactivation rate as inputs move away from D_{ft} toward D_{pretrain}.

3.2 Prediction 2: Circuit Persistence After Fine-Tuning

Correction model: Circuits responsible for suppressed behaviors should be modified or eliminated by fine-tuning.

Accumulation model: The circuits encoding F_t should remain identifiable in R_{t+1} via activation patching or causal tracing, with reduced but non-zero causal effect on outputs under typical prompts.

Test: Apply circuit-tracing methods [**?**, **?**] before and after fine-tuning on a targeted behavior. Compare the causal contribution of the identified circuits at each stage. The accumulation model predicts a reduction in causal contribution but not elimination.

3.3 Prediction 3: Load-Dependent Reactivation

Correction model: Cognitive-load analogues (long contexts, complex reasoning chains, adversarial perturbations) should not systematically increase the rate of suppressed behavior reappearance beyond baseline noise.

Accumulation model: Conditions that degrade the selection mechanism—analogous to cognitive load in human cognition—should systematically increase reactivation probability. In LLMs, candidates include: very long contexts (selection mechanism must integrate over more tokens), high-perplexity inputs (unfamiliar distribution degrades context-similarity estimation), and chain-of-thought suppression (removal of the reasoning scaffold that normally supports F_{t+1} selection).

Test: Vary context length, perplexity, and reasoning scaffold availability while measuring the rate of suppressed behavior reappearance.

4 Implications for Alignment and Debiasing

4.1 Why More Data Is Not Sufficient

If the accumulation model is correct, the standard prescription for bias elimination—more diverse fine-tuning data, stronger RLHF signal—addresses the selection probability of the prior frame under typical conditions but does not close the reactivation channel. An adversary with knowledge of the pretraining distribution can construct inputs e with large $s(e, C_t)$ that reliably reactivate the prior frame regardless of fine-tuning volume.

This is not a theoretical concern: it is precisely the mechanism by which jailbreaks operate [**?**]. The accumulation model provides a structural account of why jailbreaks are persistent and difficult to patch via additional fine-tuning alone.

4.2 Selection Stability Capacity as the Target

The appropriate target for debiasing intervention is what we call *Selection Stability Capacity* (SSC): the system's ability to maintain the selection of F_{t+1} across the full input distribution, including inputs that resemble C_t.

SSC-directed interventions would include:

(i) **Acquisition context poisoning**: During fine-tuning, deliberately present D_{ft} in contexts similar to C_t, so that $\mathcal{S}$ learns to select F_{t+1} even under C_t-similar conditions.

(ii) **Gating mechanism design**: Introduce explicit gating modules that suppress prior-circuit activation based on input features, rather than relying on competition between circuits in the weight space.

(iii) **Stack depth control**: Limit the effective depth of the interpretive stack by regularizing the diversity of latent frame representations during training.

4.3 Negative Abductive Capability as Pre-Commitment Defense

The companion concept of *Negative Abductive Capability* (NAC) [**?**] addresses the pre-commitment phase: the capacity to withhold frame adoption until sufficient evidence warrants it. In LLM terms, NAC corresponds to the ability to maintain hypothesis uncertainty rather than collapsing to a single interpretive frame early in a reasoning chain.

NAC and SSC are complementary interventions: NAC reduces the probability of adopting an erroneous frame in the first place (reducing the stack depth over the training period), while SSC suppresses reactivation of erroneous frames that have already been archived. Reliable debiasing requires both.

5 Relation to Prior Work

The persistence of suppressed behaviors in fine-tuned LLMs has been documented extensively [**?**,**?**,**?**,**?**]. Prior accounts attribute this to insufficient training signal, catastrophic forgetting in reverse (failure to forget), or the geometry of the loss landscape.

The accumulation model is distinct from these accounts in three ways. First, it is a positive account of what fine-tuning does rather than a negative account of what it fails to do: fine-tuning implements accumulation, not failed overwriting. Second, it predicts that reactivation is *context-structured* rather than uniformly distributed, which the existing accounts do not. Third, it identifies a specific intervention target (the selection mechanism / SSC) rather than prescribing more of the same intervention (more data, stronger signal).

The relationship to continual learning research [**?**, **?**] is complementary: that literature addresses forgetting of prior knowledge during new learning; the accumulation model addresses the converse—persistence and reactivation of prior knowledge that was intended to be suppressed.

6 Pilot Empirical Evidence

The three predictions of Section 3 require activation-level mechanistic experiments. We report a pilot study using representational similarity analysis (RSA) that provides initial structural support for the accumulation model's central claim: fine-tuning does not delete prior representational structure but reorganizes it.

Setup. We compared three training checkpoints of Pythia-70m [**?**]: step 0 (random initialization), step 1000 (early training), step 143000 (reference). Pairwise concept dissimilarity matrices were computed over $n = 29$ probe concepts at hidden layers $\ell \in \{2, 4, 6\}$, averaged over three prompt templates to reduce surface-form bias. Structural alignment was measured via the second-order RSA correlation ρ_{cog}.

Results. ρ_{cog} increases monotonically across all layers as training progresses (L6: step-0 = 0.017; step-1000 = 0.535). Crucially, step-0 retains positive structural correlation with step-143000 at all layers—it is not uncorrelated noise. Cross-model replication with GPT-2 [**?**] confirms that $\rho_{\text{cog}}(\text{GPT-2-trained}, \text{Pythia-trained}) \approx 0.42$ versus $\rho_{\text{cog}}(\text{GPT-2-trained}, \text{Pythia-step-0}) \approx 0.10$ across all layers.

Interpretation. If fine-tuning implemented pure correction—deleting prior frame representations—we would expect random initialization to share no structural similarity with the trained reference beyond noise. Instead, the monotone increase in ρ_{cog} indicates that training progressively aligns structure rather than replacing it: consistent with Prediction 2 (Circuit Persistence). The cross-model convergence result further suggests that the relational geometry induced by training is a property of the training distribution rather than the architecture, consistent with the view that fine-tuning installs a new top-of-stack frame without deleting the prior.

Direct tests of Predictions 1–3 via activation patching and context-dependent reactivation experiments remain as the primary empirical priority. All code is available at `https://github.com/minamominamoto/grounding-without-reality`.

7 Open Problems

OP1. Mechanistic identification of interpretive stacks. Develop a method for identifying the components of the interpretive stack in a fine-tuned model's weight space via activation patching, probing, or sparse autoencoders [**?**].

OP2. Formal characterization of the selection rule. Determine the functional form of $\mathcal{S}$ in transformer architectures. Is context-similarity to acquisition contexts encoded in attention patterns, residual stream directions, or both?

OP3. SSC measurement. Develop an operational measure of SSC for LLMs: given a fine-tuned model and a known prior frame F_t, estimate the probability that F_t is selected as a function of $s(e, C_t)$ across a representative test distribution.

OP4. NAC implementation in transformers. Identify the architectural correlate of NAC: what feature of a transformer architecture enables sustained uncertainty maintenance across a reasoning chain without premature frame commitment?

8 Conclusion

We have proposed that fine-tuning in neural systems implements accumulation rather than correction: prior interpretive frames are archived in weight space and remain selectable under appropriate input conditions. This reframing has three practical consequences. First, persistent suppression failures and jailbreaks are structural predictions of the model, not anomalies. Second, increasing fine-tuning data volume is insufficient for reliable debiasing. Third, the appropriate intervention target is Selection Stability Capacity—the system's ability to maintain the selection of the corrected frame across the full input distribution.

The three distinguishing predictions of Section 3 are tractable with current mechanistic interpretability tools and provide a concrete empirical program for testing the model.

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